

# Adaptive Body Controller (ABC) — High-Level Design

Body & Comfort Domain — AUTOSAR Classic Platform

| Field                   | Value                                       |
|-------------------------|---------------------------------------------|
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| Organization            | Synthetic Demonstration Program (fictional) |
| Security classification | Synthetic sample                            |

Revision 1.1.0 of the synthetic Adaptive Body Controller HLD. This revision contains intentional inconsistencies for evaluation purposes.

# 1 Introduction

## 1.1 Purpose

This High-Level Design (HLD) describes the software architecture of the Adaptive Body Controller (ABC). It defines software components, ports and interfaces, signals, dependencies, and functional flows. Section 3 defines the component architecture; Section 4 specifies interfaces; Section 5 lists the signal dictionary; Section 6 records dependencies; Section 7 describes functional flows.

## 1.2 Scope

The document covers the application software components of the ABC ECU, the AUTOSAR BSW services they use, and the CAN communication path. Detailed timing analysis, OS configuration, and resource files are out of scope.

## 1.3 Revision History

| Version | Date       | Author    | Changes                                                                           |
|---------|------------|-----------|-----------------------------------------------------------------------------------|
| 1.0.0   | 2026-08-14 | R. Sharma | Initial release of the Adaptive Body Controller HLD.                              |
| 1.1.0   | 2026-09-08 | R. Sharma | Architecture update: interface re-plumbing, renames, and consistency corrections. |

## 1.4 References

- [R1] AUTOSAR Classic Platform Release R22-11 documentation.  
[R2] ABC System Requirements Specification (synthetic).  
[R3] Body & Comfort CAN Communication Matrix (synthetic).

## 1.5 Terminology

| Term          | Meaning                                            |
|---------------|----------------------------------------------------|
| SWC           | Software Component (AUTOSAR application component) |
| BSW           | Basic Software                                     |
| S-R interface | Sender-Receiver interface                          |
| C-S interface | Client-Server interface                            |
| P-port        | Provided port (component provides the interface)   |
| R-port        | Required port (component requires the interface)   |

## 2 System Overview

The ABC ECU implements body and comfort functions. Application SWCs implement feature logic, the RTE connects application ports to communication services, and the BSW stack transmits signals over CAN. The BodyControlSWC (C-08) is the central arbiter: application components submit requests to it, and it releases commands only when vehicle mode (C-09) and vehicle speed (C-10) interlocks permit.

The layered view is: Application layer (C-01..C-04, C-05..C-10), RTE adapter (C-11), Services layer (C-12, C-13, C-14, C-16, C-17, C-18, C-19), and ECU abstraction (C-15, C-20).

The communication path is: application SWC -> RteAdapter -> Com -> PduR -> CanIf -> CanDriver.

### 3 Component Architecture

#### 3.1 Component Catalogue

| ID   | Component           | Type            | Layer           | Description                                                                                                      |
|------|---------------------|-----------------|-----------------|------------------------------------------------------------------------------------------------------------------|
| C-01 | DoorControlSWC      | SWC             | Application     | Controls door lock, unlock, and child-lock actuation based on switch inputs and vehicle speed constraints.       |
| C-02 | WindowLiftSWC       | SWC             | Application     | Manages one-touch window lift, anti-pinch protection, and window position learning.                              |
| C-03 | MirrorAdjustSWC     | SWC             | Application     | Controls mirror folding, heating, and position adjustment requests.                                              |
| C-04 | LightControlSWC     | SWC             | Application     | Controls interior ambient lighting and welcome light sequences.                                                  |
| C-05 | SeatAdjustSWC       | SWC             | Application     | Handles seat position and lumbar adjustment requests with memory profiles.                                       |
| C-06 | ClimateInterfaceSWC | SWC             | Application     | Bridges body domain functions to climate blower and recirculation actuators.                                     |
| C-07 | KeylessEntrySWC     | SWC             | Application     | Processes keyless authentication challenges and unlocks authorization.                                           |
| C-08 | BodyControlSWC      | SWC             | Application     | Central body domain arbiter; resolves actuator command conflicts and applies safety interlocks.                  |
| C-09 | VehicleModeMgrSWC   | SWC             | Application     | Maintains vehicle mode state (park, drive, crash) and distributes mode changes. Renamed from VehicleModeSWC.     |
| C-10 | SpeedProviderSWC    | SWC             | Application     | Provides filtered vehicle speed for anti-pinch and child-lock interlocks.                                        |
| C-11 | RteAdapter          | RTE             | RTE             | RTE adapter layer mapping SWC ports onto BSW communication services.                                             |
| C-12 | Com                 | BSW             | Service         | AUTOSAR COM module; signal packing, transmission cycles, and timeout monitoring.                                 |
| C-13 | PduR                | BSW             | Service         | PDU router; routes PDUs between COM, CanIf, and transport layers.                                                |
| C-14 | CanIf               | BSW             | Service         | CAN interface; multiplexes PDUs onto CAN controller channels.                                                    |
| C-15 | CanDriver           | BSW             | ECU-Abstraction | CAN driver for the body controller CAN channel with hardware filter configuration.                               |
| C-16 | CanSm               | BSW             | Service         | CAN state manager; coordinates bus-off recovery and channel start/stop.                                          |
| C-17 | NvM                 | BSW             | Service         | NVRAM manager; persists window pinch thresholds and seat memory profiles.                                        |
| C-18 | EcuM                | BSW             | Service         | ECU state manager; coordinates startup, shutdown, and sleep transitions.                                         |
| C-19 | DiagManager         | BSW             | Service         | Unified diagnostic services manager for UDS-based diagnostics.                                                   |
| C-20 | IoHwAb              | ECU-Abstraction | ECU-Abstraction | I/O hardware abstraction mapping physical pins to logical signals (door switches, window motors, mirror motors). |

#### 3.2 Component Details

##### 3.2.1 Component C-01: DoorControlSWC

**DoorControlSWC** (SWC, Application layer). Controls door lock, unlock, and child-lock actuation based on switch inputs and vehicle speed constraints.

| Port  | Direction | Interface      | Kind |
|-------|-----------|----------------|------|
| P-001 | provides  | DoorStatusIF   | S-R  |
| P-004 | requires  | DoorCommandIF  | C-S  |
| P-022 | requires  | VehicleModelIF | S-R  |
| P-025 | requires  | VehicleSpeedIF | S-R  |
| P-047 | requires  | IoHwDoorIF     | S-R  |

### 3.2.2 Component C-02: WindowLiftSWC

**WindowLiftSWC** (SWC, Application layer). Manages one-touch window lift, anti-pinch protection, and window position learning.

| Port  | Direction | Interface       | Kind |
|-------|-----------|-----------------|------|
| P-005 | provides  | WindowStatusIF  | S-R  |
| P-009 | requires  | WindowCommandIF | C-S  |
| P-023 | requires  | VehicleModelIF  | S-R  |
| P-026 | requires  | VehicleSpeedIF  | S-R  |
| P-040 | requires  | NvBlockIF       | C-S  |
| P-048 | requires  | IoHwDoorIF      | S-R  |

### 3.2.3 Component C-03: MirrorAdjustSWC

**MirrorAdjustSWC** (SWC, Application layer). Controls mirror folding, heating, and position adjustment requests.

| Port  | Direction | Interface        | Kind |
|-------|-----------|------------------|------|
| P-010 | provides  | MirrorStatusIF   | S-R  |
| P-013 | requires  | MirrorCommandIF  | C-S  |
| P-050 | requires  | IoHwMirrorSeatIF | S-R  |

### 3.2.4 Component C-04: LightControlSWC

**LightControlSWC** (SWC, Application layer). Controls interior ambient lighting and welcome light sequences.

| Port  | Direction | Interface       | Kind |
|-------|-----------|-----------------|------|
| P-007 | requires  | WindowStatusIF  | S-R  |
| P-014 | provides  | InteriorLightIF | S-R  |
| P-052 | requires  | IoHwLightIF     | S-R  |

### 3.2.5 Component C-05: SeatAdjustSWC

**SeatAdjustSWC** (SWC, Application layer). Handles seat position and lumbar adjustment requests with memory profiles.

This component has no explicitly assigned ports in this release; it communicates through its owning stack layer.

### 3.2.6 Component C-06: ClimateInterfaceSWC

**ClimateInterfaceSWC** (SWC, Application layer). Bridges body domain functions to climate blower and recirculation actuators.

| Port  | Direction | Interface        | Kind |
|-------|-----------|------------------|------|
| P-016 | provides  | ClimateRequestIF | C-S  |

### 3.2.7 Component C-07: KeylessEntrySWC

**KeylessEntrySWC** (SWC, Application layer). Processes keyless authentication challenges and unlocks authorization.

| Port  | Direction | Interface | Kind |
|-------|-----------|-----------|------|
| P-018 | provides  | KeyAuthIF | S-R  |

### 3.2.8 Component C-08: BodyControlSWC

**BodyControlSWC** (SWC, Application layer). Central body domain arbiter; resolves actuator command conflicts and applies safety interlocks.

| Port  | Direction | Interface        | Kind |
|-------|-----------|------------------|------|
| P-002 | requires  | DoorStatusIF     | S-R  |
| P-003 | provides  | DoorCommandIF    | C-S  |
| P-006 | requires  | WindowStatusIF   | S-R  |
| P-008 | provides  | WindowCommandIF  | C-S  |
| P-011 | requires  | MirrorStatusIF   | S-R  |
| P-012 | provides  | MirrorCommandIF  | C-S  |
| P-015 | requires  | InteriorLightIF  | S-R  |
| P-017 | requires  | ClimateRequestIF | C-S  |
| P-019 | requires  | KeyAuthIF        | S-R  |
| P-021 | requires  | VehicleModelIF   | S-R  |
| P-024 | provides  | VehicleSpeedIF   | S-R  |

### 3.2.9 Component C-09: VehicleModeMgrSWC

**VehicleModeMgrSWC** (SWC, Application layer). Maintains vehicle mode state (park, drive, crash) and distributes mode changes. Renamed from VehicleModeSWC.

| Port  | Direction | Interface      | Kind |
|-------|-----------|----------------|------|
| P-020 | provides  | VehicleModelIF | S-R  |

### 3.2.10 Component C-10: SpeedProviderSWC

**SpeedProviderSWC** (SWC, Application layer). Provides filtered vehicle speed for anti-pinch and child-lock interlocks.

This component has no explicitly assigned ports in this release; it communicates through its owning stack layer.

### 3.2.11 Component C-11: RteAdapter

**RteAdapter** (RTE, RTE layer). RTE adapter layer mapping SWC ports onto BSW communication services.

| Port  | Direction | Interface     | Kind |
|-------|-----------|---------------|------|
| P-027 | provides  | RteComIF      | S-R  |
| P-030 | requires  | RtePduIF      | C-S  |
| P-045 | requires  | DiagRequestIF | C-S  |

### 3.2.12 Component C-12: Com

**Com** (BSW, Service layer). AUTOSAR COM module; signal packing, transmission cycles, and timeout monitoring.

| Port  | Direction | Interface | Kind |
|-------|-----------|-----------|------|
| P-028 | requires  | RteComIF  | S-R  |
| P-029 | provides  | RtePduIF  | C-S  |

| Port  | Direction | Interface  | Kind |
|-------|-----------|------------|------|
| P-031 | provides  | ComPduRIF  | C-S  |
| P-043 | requires  | EcuModelIF | S-R  |

### 3.2.13 Component C-13: PduR

**PduR** (BSW, Service layer). PDU router; routes PDUs between COM, CanIf, and transport layers.

| Port  | Direction | Interface  | Kind |
|-------|-----------|------------|------|
| P-032 | requires  | ComPduRIF  | C-S  |
| P-033 | provides  | PduCanIfIF | C-S  |

### 3.2.14 Component C-14: CanIf

**CanIf** (BSW, Service layer). CAN interface; multiplexes PDUs onto CAN controller channels.

| Port  | Direction | Interface     | Kind |
|-------|-----------|---------------|------|
| P-034 | requires  | PduCanIfIF    | C-S  |
| P-035 | provides  | CanIfDriverIF | C-S  |

### 3.2.15 Component C-15: CanDriver

**CanDriver** (BSW, ECU-Abstraction layer). CAN driver for the body controller CAN channel with hardware filter configuration.

| Port  | Direction | Interface      | Kind |
|-------|-----------|----------------|------|
| P-036 | requires  | CanIfDriverIF  | C-S  |
| P-038 | requires  | CanSmControlIF | C-S  |

### 3.2.16 Component C-16: CanSm

**CanSm** (BSW, Service layer). CAN state manager; coordinates bus-off recovery and channel start/stop.

| Port  | Direction | Interface      | Kind |
|-------|-----------|----------------|------|
| P-037 | provides  | CanSmControlIF | C-S  |
| P-042 | requires  | EcuModelIF     | S-R  |

### 3.2.17 Component C-17: NvM

**NvM** (BSW, Service layer). NVRAM manager; persists window pinch thresholds and seat memory profiles.

| Port  | Direction | Interface | Kind |
|-------|-----------|-----------|------|
| P-039 | provides  | NvBlockIF | C-S  |

### 3.2.18 Component C-18: EcuM

**EcuM** (BSW, Service layer). ECU state manager; coordinates startup, shutdown, and sleep transitions.

| Port  | Direction | Interface  | Kind |
|-------|-----------|------------|------|
| P-041 | provides  | EcuModelIF | S-R  |

### 3.2.19 Component C-19: DiagManager

**DiagManager** (BSW, Service layer). Unified diagnostic services manager for UDS-based diagnostics.

| Port  | Direction | Interface     | Kind |
|-------|-----------|---------------|------|
| P-044 | provides  | DiagRequestIF | C-S  |

3.2.20 Component C-20: IoHwAb

**IoHwAb** (ECU-Abstraction, ECU-Abstraction layer). I/O hardware abstraction mapping physical pins to logical signals (door switches, window motors, mirror motors).

| Port  | Direction | Interface        | Kind |
|-------|-----------|------------------|------|
| P-046 | provides  | IoHwDoorIF       | S-R  |
| P-049 | provides  | IoHwMirrorSeatIF | S-R  |
| P-051 | provides  | IoHwLightIF      | S-R  |

## 4 Interface Specifications

Each interface below lists its provider (P-port owner) and its consumers (R-port owners). Signal definitions for each interface are given in Section 5.

### 4.1 DoorStatusIF (IF-01)

**DoorStatusIF** is a S-R interface. Provider: C-01 (DoorControlSWC). Consumers: C-08 (BodyControlSWC).

| Signal          | Datatype | Unit |
|-----------------|----------|------|
| DoorStatus      | uint8    | -    |
| ChildLockActive | boolean  | -    |

### 4.2 DoorCommandIF (IF-02)

**DoorCommandIF** is a C-S interface. Provider: C-08 (BodyControlSWC). Consumers: C-01 (DoorControlSWC).

| Signal          | Datatype | Unit |
|-----------------|----------|------|
| DoorLockRequest | uint8    | -    |

### 4.3 WindowStatusIF (IF-03)

**WindowStatusIF** is a S-R interface. Provider: C-02 (WindowLiftSWC). Consumers: C-08 (BodyControlSWC), C-04 (LightControlSWC).

| Signal             | Datatype | Unit |
|--------------------|----------|------|
| WindowPosition     | uint8    | %    |
| AntiPinchTriggered | boolean  | -    |

### 4.4 WindowCommandIF (IF-04)

**WindowCommandIF** is a C-S interface. Provider: C-08 (BodyControlSWC). Consumers: C-02 (WindowLiftSWC).

| Signal            | Datatype | Unit |
|-------------------|----------|------|
| WindowLiftCommand | sint8    | -    |

WindowCommandIF exposes two ports on the consuming side and one on the providing side (see table).

### 4.5 MirrorStatusIF (IF-05)

**MirrorStatusIF** is a S-R interface. Provider: C-03 (MirrorAdjustSWC). Consumers: C-08 (BodyControlSWC).

| Signal          | Datatype | Unit |
|-----------------|----------|------|
| MirrorFoldState | boolean  | -    |
| MirrorPosition  | uint8    | deg  |

### 4.6 MirrorCommandIF (IF-06)

**MirrorCommandIF** is a C-S interface. Provider: C-08 (BodyControlSWC). Consumers: C-03 (MirrorAdjustSWC).

| Signal              | Datatype | Unit |
|---------------------|----------|------|
| MirrorAdjustCommand | uint8    | -    |

### 4.7 InteriorLightIF (IF-07)

**InteriorLightIF** is a S-R interface. Provider: C-04 (LightControlSWC). Consumers: C-08 (BodyControlSWC).

| Signal             | Datatype | Unit |
|--------------------|----------|------|
| InteriorLightLevel | uint8    | %    |

#### 4.8 ClimateRequestIF (IF-10)

**ClimateRequestIF** is a C-S interface. Provider: C-06 (ClimateInterfaceSWC). Consumers: C-08 (BodyControlSWC).

| Signal             | Datatype | Unit |
|--------------------|----------|------|
| BlowerLevelRequest | uint8    | %    |

#### 4.9 KeyAuthIF (IF-11)

**KeyAuthIF** is a S-R interface. Provider: C-07 (KeylessEntrySWC). Consumers: C-08 (BodyControlSWC).

SG-015 (KeyAuthStatus) was removed from this interface in revision 1.1.0; the section text above still describes the authentication status signal for legacy readers.

#### 4.10 VehicleModeIF (IF-12)

**VehicleModeIF** is a S-R interface. Provider: C-09 (VehicleModeMgrSWC). Consumers: C-08 (VehicleModeSWC), C-01 (VehicleModeSWC), C-02 (VehicleModeSWC).

| Signal      | Datatype | Unit |
|-------------|----------|------|
| VehicleMode | uint8    | -    |
| CrashSignal | boolean  | -    |

#### 4.11 VehicleSpeedIF (IF-13)

**VehicleSpeedIF** is a S-R interface. Provider: C-08 (BodyControlSWC). Consumers: C-01 (DoorControlSWC), C-02 (WindowLiftSWC).

| Signal        | Datatype | Unit |
|---------------|----------|------|
| VehicleSpeed  | uint16   | km/h |
| SpeedValidity | boolean  | -    |

Note: the interface table lists BodyControlSWC as an additional provider of VehicleSpeedIF following the 1.1.0 integration review; the descriptive text retains the original SpeedProviderSWC wording.

#### 4.12 RteComIF (IF-14)

**RteComIF** is a S-R interface. Provider: C-11 (RteAdapter). Consumers: C-12 (Com).

| Signal          | Datatype | Unit |
|-----------------|----------|------|
| ComSignalBatch1 | uint8[8] | -    |

#### 4.13 RtePduIF (IF-15)

**RtePduIF** is a C-S interface. Provider: C-12 (Com). Consumers: C-11 (RteAdapter).

| Signal          | Datatype | Unit |
|-----------------|----------|------|
| ComSignalBatch2 | uint8[8] | -    |

#### 4.14 ComPduRIF (IF-16)

**ComPduRIF** is a C-S interface. Provider: C-12 (Com). Consumers: C-13 (PduR).

| Signal     | Datatype  | Unit |
|------------|-----------|------|
| PduPayload | uint8[64] | -    |

#### 4.15 PduCanIfIF (IF-17)

**PduCanIfIF** is a C-S interface. Provider: C-13 (PduR). Consumers: C-14 (CanIf).

| Signal   | Datatype | Unit |
|----------|----------|------|
| CanFrame | uint8[8] | -    |

#### 4.16 CanIfDriverIF (IF-18)

**CanIfDriverIF** is a C-S interface. Provider: C-14 (CanIf). Consumers: C-15 (CanDriver).

| Signal      | Datatype | Unit |
|-------------|----------|------|
| CanFrameRaw | uint8[8] | -    |

#### 4.17 CanSmControlIF (IF-19)

**CanSmControlIF** is a C-S interface. Provider: C-16 (CanSm). Consumers: C-15 (CanDriver).

| Signal          | Datatype | Unit |
|-----------------|----------|------|
| CanChannelState | uint8    | -    |

#### 4.18 NvBlockIF (IF-20)

**NvBlockIF** is a C-S interface. Provider: C-17 (NvM). Consumers: C-02 (WindowLiftSWC).

| Signal        | Datatype | Unit |
|---------------|----------|------|
| NvBlockStatus | uint8    | -    |

#### 4.19 EcuModelIF (IF-21)

**EcuModelIF** is a S-R interface. Provider: C-18 (EcuM). Consumers: C-16 (CanSm), C-12 (Com).

| Signal       | Datatype | Unit |
|--------------|----------|------|
| EcuModeState | uint8    | -    |

#### 4.20 DiagRequestIF (IF-22)

**DiagRequestIF** is a C-S interface. Provider: C-19 (DiagManager). Consumers: C-11 (RteAdapter).

| Signal             | Datatype   | Unit |
|--------------------|------------|------|
| DiagRequestPayload | uint8[255] | -    |

#### 4.21 IoHwDoorIF (IF-23)

**IoHwDoorIF** is a S-R interface. Provider: C-20 (IoHwAb). Consumers: C-01 (DoorControlSWC), C-02 (WindowLiftSWC).

| Signal             | Datatype | Unit |
|--------------------|----------|------|
| DoorSwitchRaw      | uint8    | -    |
| WindowMotorCurrent | uint16   | mA   |

#### 4.22 IoHwMirrorSeatIF (IF-24)

**IoHwMirrorSeatIF** is a S-R interface. Provider: C-20 (IoHwAb). Consumers: C-03 (MirrorAdjustSWC).

| Signal              | Datatype | Unit |
|---------------------|----------|------|
| MirrorMotorFeedback | uint16   | mA   |
| SeatMotorFeedback   | uint16   | mA   |

#### 4.23 IoHwLightIF (IF-25)

**IoHwLightIF** is a S-R interface. Provider: C-20 (IoHwAb). Consumers: C-04 (LightControlSWC).

| Signal               | Datatype | Unit |
|----------------------|----------|------|
| LightSwitchRaw       | uint8    | -    |
| AmbientLightFeedback | uint8    | %    |

## 5 Signal Dictionary

The signal dictionary lists all signals carried by the interfaces of Section 4, including datatype and physical unit. The VehicleSpeed signal (SG-018) is filtered by the SpeedProviderSWC before distribution.

| Signal ID | Signal               | Datatype   | Unit | Interface        |
|-----------|----------------------|------------|------|------------------|
| SG-001    | DoorStatus           | uint8      | -    | DoorStatusIF     |
| SG-002    | ChildLockActive      | boolean    | -    | DoorStatusIF     |
| SG-003    | DoorLockRequest      | uint8      | -    | DoorCommandIF    |
| SG-004    | WindowPosition       | uint8      | %    | WindowStatusIF   |
| SG-005    | AntiPinchTriggered   | boolean    | -    | WindowStatusIF   |
| SG-006    | WindowLiftCommand    | sint8      | -    | WindowCommandIF  |
| SG-007    | MirrorFoldState      | boolean    | -    | MirrorStatusIF   |
| SG-008    | MirrorPosition       | uint8      | deg  | MirrorStatusIF   |
| SG-009    | MirrorAdjustCommand  | uint8      | -    | MirrorCommandIF  |
| SG-010    | InteriorLightLevel   | uint8      | %    | InteriorLightIF  |
| SG-014    | BlowerLevelRequest   | uint8      | %    | ClimateRequestIF |
| SG-016    | VehicleMode          | uint8      | -    | VehicleModelIF   |
| SG-017    | CrashSignal          | boolean    | -    | VehicleModelIF   |
| SG-018    | VehicleSpeed         | uint16     | km/h | VehicleSpeedIF   |
| SG-019    | SpeedValidity        | boolean    | -    | VehicleSpeedIF   |
| SG-020    | ComSignalBatch1      | uint8[8]   | -    | RteComIF         |
| SG-021    | ComSignalBatch2      | uint8[8]   | -    | RtePduIF         |
| SG-022    | PduPayload           | uint8[64]  | -    | ComPduRIF        |
| SG-023    | CanFrame             | uint8[8]   | -    | PduCanflIF       |
| SG-024    | CanFrameRaw          | uint8[8]   | -    | CanIfDriverIF    |
| SG-025    | CanChannelState      | uint8      | -    | CanSmControlIF   |
| SG-026    | NvBlockStatus        | uint8      | -    | NvBlockIF        |
| SG-027    | EcuModeState         | uint8      | -    | EcuModelIF       |
| SG-028    | DiagRequestPayload   | uint8[255] | -    | DiagRequestIF    |
| SG-029    | DoorSwitchRaw        | uint8      | -    | IoHwDoorIF       |
| SG-030    | WindowMotorCurrent   | uint16     | mA   | IoHwDoorIF       |
| SG-031    | MirrorMotorFeedback  | uint16     | mA   | IoHwMirrorSeatIF |
| SG-032    | SeatMotorFeedback    | uint16     | mA   | IoHwMirrorSeatIF |
| SG-033    | LightSwitchRaw       | uint8      | -    | IoHwLightIF      |
| SG-034    | AmbientLightFeedback | uint8      | %    | IoHwLightIF      |

## 6 Dependencies

### 6.1 Dependency Overview

Dependencies record which components require services or commands from other components. Every dependency in this section is realized by an interface defined in Section 4.

| Dep. ID | Source                     | Target                   | Type       |
|---------|----------------------------|--------------------------|------------|
| DEP-01  | C-01 (DoorControlSWC)      | C-08 (BodyControlSWC)    | requires   |
| DEP-02  | C-02 (WindowLiftSWC)       | C-08 (BodyControlSWC)    | requires   |
| DEP-03  | C-03 (MirrorAdjustSWC)     | C-08 (BodyControlSWC)    | requires   |
| DEP-05  | C-06 (ClimateInterfaceSWC) | C-08 (BodyControlSWC)    | requires   |
| DEP-06  | C-08 (BodyControlSWC)      | C-09 (VehicleModeMgrSWC) | requires   |
| DEP-07  | C-08 (BodyControlSWC)      | C-10 (SpeedProviderSWC)  | requires   |
| DEP-08  | C-02 (WindowLiftSWC)       | C-10 (SpeedProviderSWC)  | requires   |
| DEP-09  | C-01 (DoorControlSWC)      | C-07 (KeylessEntrySWC)   | requires   |
| DEP-10  | C-11 (RteAdapter)          | C-12 (Com)               | depends_on |
| DEP-11  | C-12 (Com)                 | C-13 (PduR)              | depends_on |
| DEP-12  | C-13 (PduR)                | C-14 (CanIf)             | depends_on |
| DEP-13  | C-14 (CanIf)               | C-15 (CanDriver)         | depends_on |
| DEP-14  | C-16 (CanSm)               | C-15 (CanDriver)         | requires   |
| DEP-15  | C-02 (WindowLiftSWC)       | C-17 (NmM)               | requires   |
| DEP-17  | C-18 (EcuM)                | C-16 (CanSm)             | requires   |
| DEP-18  | C-18 (EcuM)                | C-12 (Com)               | requires   |
| DEP-20  | C-01 (DoorControlSWC)      | C-20 (IoHwAb)            | requires   |
| DEP-21  | C-02 (WindowLiftSWC)       | C-20 (IoHwAb)            | requires   |
| DEP-22  | C-03 (MirrorAdjustSWC)     | C-20 (IoHwAb)            | requires   |
| DEP-23  | C-04 (LightControlSWC)     | C-20 (IoHwAb)            | requires   |

### 6.2 Dependency Details

#### 6.2.1 DEP-01

The DoorControlSWC requires door lock and unlock commands from the BodyControlSWC.

#### 6.2.2 DEP-02

The WindowLiftSWC requires window lift commands from the BodyControlSWC.

#### 6.2.3 DEP-03

The MirrorAdjustSWC requires mirror adjustment commands from the BodyControlSWC.

#### 6.2.4 DEP-05

The ClimateInterfaceSWC requires blower control authorization from the BodyControlSWC.

Note (1.1.0): ClimateInterfaceSWC does not require any interface from the BodyControlSWC; blower requests are issued autonomously.

### 6.2.5 DEP-06

The BodyControlSWC requires vehicle mode information from the VehicleModeSWC before releasing actuator commands.

### 6.2.6 DEP-07

The BodyControlSWC requires filtered vehicle speed from the SpeedProviderSWC for anti-pinch interlocks.

### 6.2.7 DEP-08

The WindowLiftSWC requires vehicle speed to qualify one-touch window operation.

### 6.2.8 DEP-09

The DoorControlSWC requires keyless authorization status from the KeylessEntrySWC.

### 6.2.9 DEP-10

The RteAdapter depends on the COM module for signal transmission.

### 6.2.10 DEP-11

The COM module depends on the PduR for PDU routing.

### 6.2.11 DEP-12

The PduR depends on the CanIf for CAN transmission.

### 6.2.12 DEP-13

The CanIf depends on the CanDriver for hardware access.

### 6.2.13 DEP-14

The CanSm requires bus state control of the CanDriver.

### 6.2.14 DEP-15

The WindowLiftSWC requires NVRAM persistence of anti-pinch thresholds from the NvM.

### 6.2.15 DEP-17

The EcuM requires communication channel control from the CanSm during shutdown.

### 6.2.16 DEP-18

The EcuM requires COM transmission suspension during shutdown.

### 6.2.17 DEP-20

The DoorControlSWC requires raw door switch inputs from the IoHwAb.

### 6.2.18 DEP-21

The WindowLiftSWC requires window motor current feedback from the IoHwAb.

### 6.2.19 DEP-22

The MirrorAdjustSWC requires mirror motor feedback from the IoHwAb.

#### **6.2.20 DEP-23**

The LightControlSWC requires ambient light feedback from the IoHwAb.

## 7 Functional Flows

Functional flows describe end-to-end behaviour across components. Flow steps reference the components of Section 3.

### 7.1 Door Lock Actuation (FL-1)

**Trigger:** Driver operates door lock switch

**Flow:** C-20 (IoHwAb) → C-01 (DoorControlSWC) → C-08 (BodyControlSWC) → C-11 (RteAdapter) → C-12 (Com) → C-13 (PduR) → C-14 (CanIf) → C-15 (CanDriver)

Raw switch input is debounced in IoHwAb, evaluated by DoorControlSWC, arbitrated by BodyControlSWC, and the resulting command is transmitted over CAN.

### 7.2 One-Touch Window Lift with Anti-Pinch (FL-2)

**Trigger:** Driver triggers one-touch window lift

**Flow:** C-20 (IoHwAb) → C-02 (WindowLiftSWC) → C-10 (SpeedProviderSWC) → C-08 (BodyControlSWC) → C-11 (RteAdapter) → C-12 (Com) → C-13 (PduR) → C-14 (CanIf) → C-15 (CanDriver)

Motor current from IoHwAb is monitored by WindowLiftSWC; vehicle speed from SpeedProviderSWC qualifies the command; BodyControlSWC arbitrates and forwards the request over CAN.

### 7.3 Keyless Unlock (FL-3)

**Trigger:** Authenticated key approach detected

**Flow:** C-07 (KeylessEntrySWC) → C-08 (BodyControlSWC) → C-01 (DoorControlSWC) → C-20 (IoHwAb)

KeylessEntrySWC publishes the authentication result; BodyControlSWC authorizes the unlock; DoorControlSWC actuates via IoHwAb.

### 7.4 Bus-Off Recovery (FL-5)

**Trigger:** CAN controller enters bus-off state

**Flow:** C-15 (CanDriver) → C-16 (CanSm) → C-18 (EcuM)

CanDriver reports the fault, CanSm executes the recovery sequence, and EcuM is informed for mode bookkeeping.

## 8 Assumptions and Open Points

### 8.1 Assumptions

- A1. The CAN bus operates at 500 kbit/s.
- A2. The NvM stores up to three seat memory profiles per user.
- A3. Anti-pinch protection is mandatory for all windows when vehicle speed is below 5 km/h.

### 8.2 Open Points

OP-2: Welcome-light sequencing depends on vehicle mode distribution (IF-12). WindowStatusIF is additionally monitored by LightControlSWC since 1.1.0 (new consumer).